Mathematical Manipulation as Structured Transformation: Toward an Operational Account of Mathematical Practice

Kevin T.N.*

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Abstract

This paper develops an operational account of mathematical practice. The central thesis is that mathematical manipulation is not merely a transparent vehicle for accessing pre-given structures. Rather, transformations on representations are themselves structured, invariant-constrained operations that contribute constitutively to mathematical understanding. The argument proceeds in four steps. First, I distinguish object-level, representation-level, operational, and cognitive-enactive structure. Second, I show why a purely object-centered or result-centered view misses an essential feature of actual mathematical work. Third, I formalize mathematical manipulation as a rule-governed transformation between representations under invariant constraints. Finally, I illustrate the proposal by means of two case studies: change of basis in linear algebra and proof transformations in mathematical reasoning. The resulting picture is compatible with structuralist insights in the philosophy of mathematics, but it shifts attention from static structures alone to the structured paths by which mathematicians traverse them. A key contribution of the paper is to clarify how this proposal differs from a merely category-theoretic emphasis on morphisms: formal mappings characterize admissible structural relations, but do not by themselves exhaust the epistemic role of selecting, sequencing, and enacting transformations in mathematical practice.

Keywords: philosophy of mathematics; mathematical practice; structuralism; category theory; representation; invariance; proof

1. Introduction

Much of the philosophy of mathematics has been organized around questions of ontology, truth, and objectivity. What are mathematical objects? In virtue of what are mathematical statements true? How is mathematical knowledge possible? Structuralist approaches have usefully shifted attention away from independent objects toward patterns, places in structures, and the relations that hold among them (Resnik 1997; Shapiro 1997). Likewise, category-theoretic perspectives have emphasized the primacy of morphisms, functoriality, and systems of relations (Lane 1998; Awodey 2004; Awodey 2010).

Yet even these important shifts often leave underdescribed a feature that is central to actual mathematical work: mathematicians do not merely contemplate structures. They transform formulas, rewrite expressions, change coordinates, translate between representations, reorganize proofs, and reconfigure inferential pathways. In short, mathematics is not only a body of structures and results; it is also a disciplined practice of structured transformation (Lakatos 1976; Mancosu 2008; Corfield 2003).

*jkdkr2439@gmail.com

The present paper argues for the following thesis: in mathematical practice, transformations on representations are not merely instrumental steps that disappear once the final result has been reached. They are themselves structured operations that help constitute mathematical understanding. To know a mathematical object is not only to know what it is or what follows from it; it is also, in many cases, to know how it may be traversed through admissible transformations that preserve relevant invariants while reorganizing what is salient.

The point is not that mathematical objects dissolve into human psychology, nor that formal structure should be replaced by a sociology of calculation. The claim is narrower and stronger: within mathematical practice, the operational layer has genuine epistemic and structural significance. It is not transparent. Treating it as transparent obscures how mathematical knowledge is actually produced, stabilized, and communicated.

The paper is organized as follows. Section 2 situates the proposal within structuralism, category theory, and the philosophy of mathematical practice. Section 3 introduces a four-level distinction needed to avoid conceptual slippage. Section 4 states the core thesis. Section 5 offers a minimal formal frame. Sections 6 and 7 develop two case studies. Section 8 then addresses the most likely objection, namely that category theory already captures everything important here through its account of morphisms. Section 9 considers broader objections and clarifies the limits of the view.

2. Background and Problem Setting

Structuralism in the philosophy of mathematics characteristically holds that mathematics concerns structures rather than self-subsisting objects with intrinsic natures (Resnik 1997; Shapiro 1997). On this view, what matters are the positions, relations, and patterns that persist across appropriate systems. Category theory radicalizes some of these intuitions by foregrounding maps and transformations. A mathematical domain can often be understood through the morphisms that relate its objects, the functors that connect categories, and the invariants that survive translation (Lane 1998; Awodey 2004; Awodey 2010).

At the same time, work in the philosophy of mathematical practice has argued that philosophical reflection should attend more closely to what mathematicians actually do: experiment, diagram, conjecture, revise, generalize, and prove (Mancosu 2008; Corfield 2003). Lakatos's classic analysis of proof and refutation remains especially important here, because it shows that mathematical growth is often inseparable from the transformations by which a theorem, a counterexample, or a proof schema is reworked (Lakatos 1976). Later work on proof, visualization, and concept formation likewise emphasizes that mathematical understanding often depends on routes of reorganization, not only on final statements (Giaquinto 2007; Carter 2019; Tappenden 2005).

The gap addressed in this paper lies at the intersection of these traditions. Structuralism tells us to care about structure. Category theory teaches us to care about morphisms. Philosophy of practice reminds us that mathematics is enacted. What remains underdeveloped is a unified account of mathematical manipulation as a structured and epistemically constitutive component of mathematical practice.

This gap becomes visible whenever mathematical work is described too quickly as if it moved directly from theorem to theorem, or from object to object, without attending to the concrete pathways through which intelligibility is produced. But in practice, much of what mathematicians value lies precisely in the shape of those pathways: whether a derivation is natural or forced, whether a change of coordinates reveals hidden symmetry, whether a proof method generalizes, whether a normalization procedure exposes a deep invariant, and whether a transformation

creates new connections rather than merely certifying an old one.

3. A Four-Level Distinction

To avoid category mistakes, it is helpful to distinguish four levels.

3.1. Object-Level Structure

This is the level of mathematical objects or structures themselves: groups, vector spaces, topological spaces, categories, or proof targets. It is the level most often foregrounded in standard ontology-driven discussions.

3.2. Representation-Level Structure

This is the level at which objects are rendered in formulas, matrices, diagrams, symbolic strings, coordinate systems, proof notations, or computational encodings. A single object-level structure may admit many non-equivalent representations, each of which makes certain features visible and others obscure.

3.3. Operational Structure

This is the level of admissible transformations between representations. Examples include substitution, factorization, change of basis, normalization, equivalence transformations, duality maps, and proof transformations. These operations are not arbitrary. They are rule-governed and constrained by what must be preserved.

3.4. Cognitive-Enactive Structure

This is the level at which agents recognize, select, sequence, and execute transformations. It includes judgments of relevance, salience, elegance, convenience, explanatory gain, and strategic fruitfulness. This level does not reduce mathematics to psychology, but it does register the fact that mathematical understanding is enacted through practice.

The distinction is methodological rather than metaphysical. Its purpose is to show that several debates are easily confused if one slides too quickly between levels. An object-level invariant is not the same thing as a representation-level convenience. A formally available operation is not the same thing as a salient or productive move in practice. A proof may exist in principle while remaining opaque until a certain route of transformation makes it intelligible. The paper's central claim depends on keeping these levels apart.

4. Core Thesis

The central claim of this paper concerns the relation between the representation level and the operational level.

Claim 1. *In mathematical practice, the transformation from one representation to another is not merely a neutral conduit. It is a structured operation that contributes to what is known, how it is known, and which aspects of a mathematical object become intelligible.*

This claim should be understood neither as a denial of objectivity nor as a full-blown anti-realist thesis. The target is a simpler assumption: that the epistemic work is exhausted by the

object and the final result, while the transformation that mediates between them is philosophically negligible. That assumption is false.

To see why, note that mathematical transformations are typically evaluated along several dimensions at once: correctness, admissibility, preservation of invariant content, inferential utility, explanatory power, and strategic fruitfulness. A proof by normalization is not epistemically identical to a proof by contradiction simply because both establish the same theorem. Likewise, two coordinate systems may present the same linear map while affording very different insight into its structure.

The point can be stated more sharply. In many settings, the difference between merely having an equivalent representation and actually understanding what that equivalence affords is a difference in operation. One may know that two forms are related without yet knowing how to move between them in a way that reveals a mechanism, exposes a symmetry, or opens a path to generalization. Operational structure therefore belongs to mathematical knowledge not as a dispensable aid, but as one of its modes of articulation.

5. Toward a Minimal Formal Frame

The proposal can be expressed schematically. Let R_1 and R_2 be two representations of a mathematical structure S, and let

$$T : R_1 \longrightarrow R_2$$

be an admissible transformation. Let $\mathcal{I}$ denote the family of invariants or structure-preserving constraints relevant to the context.

Then an operational account treats mathematical understanding as depending not only on the relation

$$R_1 \sim_{\mathcal{I}} R_2,$$

but also on the form and role of T itself.

Definition 1. *An* operational transformation *is a rule-governed map between mathematical representations such that:*

(i) it is admissible within a given practice or theory;

(ii) it preserves a contextually relevant family of invariants;

(iii) it reorganizes what is inferentially or structurally salient; and

(iv) it can therefore contribute to mathematical understanding, and not merely to syntactic passage.

This account shifts emphasis from outcomes alone to pathways under constraint. The key point is not simply that one may move from R_1 to R_2, but that the particular transformation used has epistemic shape. Some transformations expose symmetry. Others isolate conserved quantities. Others simplify proof search or reveal hidden equivalence classes. In each case, the operation is part of the structured content of the practice.

The point may be contrasted with a thin reading of equivalence. To say that two expressions are equivalent is to say something important. But it does not yet say enough. It does not specify which route of transformation makes the equivalence usable, which intermediate representations become available, which invariants are most illuminating, or why one route becomes canonical in a field while another remains technically correct but epistemically sterile.

6. Case Study I: Change of Basis

Linear algebra provides a clean test case. Consider a linear map $L : V \to V$, where V is a finite-dimensional vector space. Relative to one basis, L may be represented by a matrix A; relative to another basis, by a matrix B, with

$$B = P^{-1}AP.$$

Here, the object-level map is unchanged, but the representation has shifted. The operation of changing basis is governed by a clear rule and preserves important invariants, such as determinant, trace, rank, eigenvalues, and Jordan type where appropriate.

A purely result-centered view might say that this is merely a change in notation. But this misses the epistemic significance of the operation. The choice of basis may render diagonal structure visible, separate invariant subspaces, simplify dynamics, or reveal canonical forms. The transformation is not transparent: it changes what can be seen, proved quickly, or understood well.

This is precisely the sense in which operational structure matters. The move from A to B is not a disposable bridge between equivalent descriptions. It is often the mathematically decisive step by which the structure becomes intelligible. The path itself has explanatory content.

It is useful to distinguish two claims that are often conflated. The first is formal: A and B lie in the same similarity class. The second is epistemic: a particular change of basis can reveal a decomposition, suggest a proof, compress a computation, or make an invariant geometrically visible. The formal fact alone does not determine which basis is salient, natural, pedagogically fruitful, or research-generating. Those questions arise only once the operation is situated within practice.

Indeed, many mathematically trained agents can state that diagonalizable matrices are similar to diagonal matrices without thereby possessing the same level of understanding as someone who knows how to construct the relevant basis, when to seek it, and why its existence matters. In this sense, understanding is not exhausted by equivalence-class membership. It also includes command over admissible transformations that make such equivalences intelligible and useful.

7. Case Study II: Proof Transformations

Proofs provide an even stronger example. Suppose a theorem admits two proofs: one via contradiction and another via constructive reduction. At the level of truth value, both establish the same proposition. Yet they can differ dramatically in mathematical significance.

Lakatos showed that mathematical development often proceeds not by linear accumulation of finished truths, but by revision of proofs, reclassification of counterexamples, and restructuring of the space of acceptable inferences (Lakatos 1976). A proof transformation can expose hidden assumptions, sharpen definitions, or generate new methods that extend beyond the theorem initially proved.

This suggests that proof paths are not epistemically inert. The shape of a proof—which lemmas it uses, which transformations it licenses, which invariants it preserves, which analogies it unlocks—belongs to the structure of mathematical understanding. That is why mathematicians value not only that a theorem is true, but also how it is proved.

An operational account therefore has a natural home in proof theory and proof practice. The endpoint matters, but the route is not philosophically negligible. The route itself can carry mathematical content.

This becomes especially clear when different proofs support different futures for the subject. One proof may be local and non-transferable, while another reveals a reusable technique. One proof may certify truth with minimal insight, while another reorganizes the surrounding conceptual space. One proof may depend on a representation whose significance ends with the theorem at hand, while another may create a bridge to neighboring domains. These are not merely aesthetic differences. They track distinct epistemic roles for transformation.

For that reason, proof transformation is not an accidental feature of mathematical culture. It is one of the ways mathematics stores, transmits, and amplifies understanding. If two proofs establish the same result but only one reveals a method that can be exported, abstracted, or generalized, then the difference lies not in truth conditions alone but in the operational shape of the reasoning.

8. Relation to Structuralism and Category Theory

This section addresses the most serious challenge to the present proposal. If category theory already emphasizes morphisms, maps, and transformations, why is an additional account of mathematical manipulation needed at all? Is the present view simply a more rhetorical restatement of a point already captured by the morphism-first perspective?

The answer is no, but the reason must be stated carefully. The point is not that category theory neglects transformations. Category theory is one of the deepest formal languages we possess for describing them. Morphisms, functors, natural transformations, and universal properties all testify to the fact that mathematics cannot be reduced to isolated objects (Lane 1998; Awodey 2010). In that sense, category theory is an ally of the present paper, not its opponent.

The difference lies elsewhere. Category theory characterizes transformations at the formal level. It tells us what counts as a structure-preserving map, how such maps compose, which properties are invariant under translation, and how mathematical domains are related in systematic ways. But a formal characterization of admissible mappings does not by itself exhaust the epistemic role of transformations in mathematical practice.

What category theory captures, in the first instance, is the existence and structure of morphic relations. What the present paper adds is an account of how transformations function when they are selected, sequenced, and enacted within mathematical work. Formal morphisms and enacted operations are related, but they are not identical objects of analysis. The former belong primarily to the formal articulation of a theory. The latter belong to the epistemic life of mathematics as practiced.

This difference can be sharpened through the notion of *epistemic underdetermination by formal equivalence*. Suppose that several formally legitimate transformations connect two representations of the same structure. The formal fact that such transformations exist does not determine which one will be illuminating, strategically fruitful, pedagogically natural, or methodologically generative. Yet these are precisely the questions that matter in actual mathematical inquiry.

8.1. A Contrast Case

Consider again change of basis in linear algebra. At the formal level, similarity transformations capture the fact that two matrices may represent the same linear operator. This belongs squarely within standard mathematics and can be described in structural terms without difficulty. But that formal relation alone does not settle which basis matters in practice.

One basis may diagonalize the operator and thereby reveal spectral structure. Another may align with a geometric symmetry. A third may be computationally convenient for iteration or

numerical approximation. A fourth may make an invariant subspace decomposition obvious. These choices are not reducible to the bare existence of an isomorphism class. They concern the epistemic yield of particular transformations as enacted within particular contexts.

In this sense, formal equivalence does not determine epistemic salience. Two routes may be equally legitimate as mappings while being markedly unequal as vehicles of understanding. One route may show why something is true. Another may only show that it is true. One route may generalize. Another may dead-end. One may expose a deep invariant. Another may leave the same invariant implicit and unusable.

Proof transformations provide an even stronger contrast. If a theorem admits multiple proofs, category-theoretic language may capture formal relations among structures involved in those proofs, but it does not by itself explain why one proof becomes canonical, why one proof reorganizes a field, or why one proof route yields a new research program while another remains merely confirmatory. Those differences concern the inferential and explanatory role of the enacted path.

The upshot is not that category theory is incomplete in some crude sense. Rather, it is that the philosophy of mathematical practice asks a different question. Category theory asks, in part, what the formal relations are. The present paper asks what it means to know through traversing such relations under conditions of inquiry. Morphisms characterize permissible structural relations; they do not by themselves determine the cognitive salience, heuristic value, or explanatory force of their enactment in practice.

8.2. Operational Structuralism

The position defended here may therefore be described as a limited form of *operational structuralism*. It retains the structuralist insight that mathematics concerns relations, patterns, and invariants, while insisting that these are often accessible only through structured operations whose epistemic role deserves explicit attention.

Operational structuralism is not an attempt to replace formal mathematics with phenomenology. It is a claim about philosophical adequacy. A satisfactory philosophy of mathematics should account not only for structures and maps as formal entities, but also for the way mathematical understanding depends on pathways of transformation that are strategically and inferentially inhabited.

9. Relation to Wittgenstein and Mathematical Practice

There is also a limited affinity here with Wittgensteinian themes. Wittgenstein's attention to use warns against treating symbols as self-sufficient bearers of meaning (Wittgenstein 2009; Wittgenstein 1978). The present paper does not reduce mathematical content to use, but it does insist that the significance of a mathematical representation cannot be understood independently of the admissible transformations through which it functions within practice.

This is also why the paper aligns naturally with philosophy of mathematical practice. The proposal does not say that mathematics is merely social, psychological, or historical. It says that what counts as understanding in mathematics is inseparable from the operations by which mathematicians move, compare, simplify, stabilize, and reorganize representations. Once this is granted, the philosophy of mathematics must take practice seriously not as a sociological afterthought, but as one site at which structure is made epistemically available.

10. Objections and Replies

10.1. Objection 1: The View Collapses into Psychology

One might worry that emphasizing enacted operations collapses mathematics into human psychology. This is not the claim. The proposal concerns the structure of mathematical practice, not the metaphysical dependence of mathematics on particular minds. Operational structure can be objective, sharable, and teachable, even though it is enacted.

Moreover, the enacted dimension matters precisely because mathematics is not a sequence of arbitrary mental events. It is a rule-governed and publicly criticizable practice. The present account therefore does not psychologize mathematics; it adds a layer of practice-sensitive analysis to accounts that would otherwise remain too thin.

10.2. Objection 2: This Is Already in Category Theory

Category theory indeed foregrounds morphisms. But the present thesis concerns the epistemic and practical significance of transformations as they function in mathematical work. Formal morphisms and enacted operations are related, but not identical objects of analysis. The contrast cases above are intended to show that sameness of formal relation does not entail sameness of epistemic role.

10.3. Objection 3: Results Matter More Than Routes

Results certainly matter. But mathematical culture itself distinguishes elegant proofs from brute-force proofs, natural transformations from ad hoc calculations, illuminating changes of representation from merely cumbersome ones. Such distinctions make sense only if routes carry epistemic weight.

A result-centered account can acknowledge that routes are useful, yet still treat them as philosophically secondary. The present paper argues that this is too weak. Routes matter not merely because they help humans reach results, but because they help constitute what counts as understanding the result.

10.4. Objection 4: The View Overstates Practice

A further concern is that practice-based language may be suitable for pedagogy or discovery, but not for the philosophical analysis of mathematics proper. This objection presupposes too sharp a split between finished mathematics and mathematical activity. Many of the features by which mathematics is recognized as deep, elegant, or fruitful are inseparable from the operations by which results are produced and connected. Practice is not external decoration added to an otherwise complete formal core. It is one of the media in which mathematical intelligibility is formed.

11. Conclusion

The philosophy of mathematics should not treat mathematical manipulation as a transparent intermediary between fixed structures and final results. Transformations on representations are rule-governed, invariant-constrained, and often epistemically decisive. To understand mathematics as practiced is therefore to attend not only to objects, truths, and structures, but also to the structured operations by which representations are traversed, reorganized, and made intelligible.

The larger implication is modest but important. If mathematical knowledge is partly constituted through operational structure, then a complete philosophy of mathematics must include an account of the pathways of transformation, and not merely the termini. That is the sense in which mathematical manipulation is not transparent.

The paper's main claim is therefore not that formal structure is insufficient, but that formal structure alone does not capture the whole epistemic reality of mathematics as practiced. Mathematical understanding lives not only in objects and propositions, but also in admissible routes of transformation under invariant constraint. That is the space in which operations become philosophically visible.

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